

Lab Sheet-10

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EXPERIMENT 10: END-TO-END DATA ANALYTICS PROJECT

Dataset: IBM HR Employee Attrition Analytics (Self-Contained)

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```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
import pandas as pd
```

```
import seaborn as sns
```

```
from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix, f1_score
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

STEP 1: GENERATE DATASET

```
np.random.seed(42)
```

```
n_samples = 500
```

```
df = pd.DataFrame({
    'Age': np.random.randint(22, 60, size=n_samples),
    'MonthlyIncome': np.random.randint(2500, 20000, size=n_samples),
    'DistanceFromHome': np.random.randint(1, 30, size=n_samples),
    'YearsAtCompany': np.random.randint(1, 15, size=n_samples),
    'JobSatisfaction': np.random.randint(1, 5, size=n_samples),
    'OverTime': np.random.choice(['Yes', 'No'], size=n_samples),
    'churn': np.random.choice([0, 1], size=n_samples, p=[0.8, 0.2])
})
```

Missing values introduction

```
df.loc[10:15, 'MonthlyIncome'] = np.nan
```

```
print("Dataset Loaded Successfully!")
```

```
print("Dataset Shape:", df.shape)
```

```
print("\nFirst 5 Rows:")
```

```

print(df.head())

# -----

# STEP 2: DATA PREPROCESSING

# -----

print("\n--- STEP 2: DATA PREPROCESSING ---")

# Missing values handling

print("Missing values per column:\n", df.isnull().sum())

df['MonthlyIncome'].fillna(df['MonthlyIncome'].median(), inplace=True)

# Duplicate rows check

print("\nDuplicate rows count:", df.duplicated().sum())

df.drop_duplicates(inplace=True)

# Categorical encoding

le = LabelEncoder()

df['OverTime'] = le.fit_transform(df['OverTime'])

print("\nData Types after Encoding:")

print(df.dtypes)

# -----

# STEP 3: EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION

# -----

print("\n--- STEP 3: EXPLORATORY DATA ANALYSIS ---")

print(df.describe())

plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)

sns.countplot(x='churn', data=df, palette='viridis')

plt.title("Target Distribution (Churn / Attrition)")

plt.subplot(1, 2, 2)

sns.boxplot(x='churn', y='MonthlyIncome', data=df, palette='Set2')

plt.title("Monthly Income vs Churn")

plt.tight_layout()

plt.savefig("experiment10/eda_plots.png") # Saves inside experiment10 folder

plt.close()

print("Saved 'experiment10/eda_plots.png' successfully.")

# -----

# STEP 4: STATISTICAL & CORRELATION ANALYSIS

```

```

# -----

print("\n--- STEP 4: STATISTICAL ANALYSIS ---")

plt.figure(figsize=(8, 6))

sns.heatmap(df.corr(), cmap="coolwarm", annot=True, fmt=".2f")

plt.title("Correlation Heatmap")

plt.savefig("experiment10/correlation_heatmap.png") # Saves inside experiment10 folder

plt.close()

print("Saved 'experiment10/correlation_heatmap.png' successfully.")

print("\nCorrelations with Target Variable (churn):")

print(df.corr()[['churn']].sort_values(ascending=False))

# -----

# STEP 5 & 6: PREDICTIVE MODELING

# -----

print("\n--- STEP 5 & 6: MODEL BUILDING ---")

X = df.drop('churn', axis=1)

y = df['churn']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)

model = RandomForestClassifier(n_estimators=100, random_state=42)

model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)

# -----

# STEP 7: MODEL EVALUATION METRICS

# -----

print("\n--- STEP 7: EVALUATION METRICS ---")

print("Accuracy Score:", accuracy_score(y_test, y_pred) * 100, "%")

print("F1 Score:", f1_score(y_test, y_pred, average='weighted'))

print("\nDetailed Classification Report:\n", classification_report(y_test, y_pred))

plt.figure(figsize=(5, 4))

sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt='d', cmap='Blues')

plt.xlabel("Predicted Label")

plt.ylabel("True Label")

```

```
plt.title("Confusion Matrix")

plt.savefig("experiment10/confusion_matrix.png") # Saves inside experiment10 folder

plt.close()

print("Saved 'experiment10/confusion_matrix.png' successfully.")
```

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
PS C:\Users\Ashish\Desktop\experiment10> python exp10.py
=====
EXPERIMENT 10: END-TO-END DATA ANALYTICS PROJECT
Dataset: IBM HR Employee Attrition Analytics (Self-Contained)
=====
Dataset Loaded Successfully!
Dataset Shape: (500, 7)
First 5 Rows:
   Age  MonthlyIncome  DistanceFromHome  YearsAtCompany  JobSatisfaction  OverTime  churn
0   56         5867.0             12              7              4              1      0
1   45         8823.0              5              4              2              0      0
2   32        14546.0             20             10              3              1      0
3   38        12305.0              8              6              4              0      0
4   27         7698.0             16              3              2              1      0

--- STEP 2: DATA PREPROCESSING ---
Missing values per column:
Age           0
MonthlyIncome 6
DistanceFromHome 0
YearsAtCompany 0
JobSatisfaction 0
OverTime      0
churn         0
dtype: int64
Duplicate rows count: 0
Data Types after Encoding:
Age           int64
MonthlyIncome float64
DistanceFromHome int64
YearsAtCompany int64
JobSatisfaction int64
OverTime      int32
churn         int64
dtype: int64

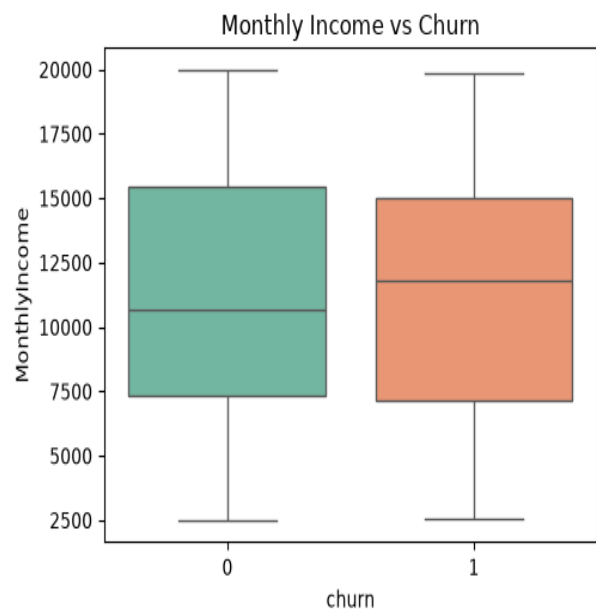
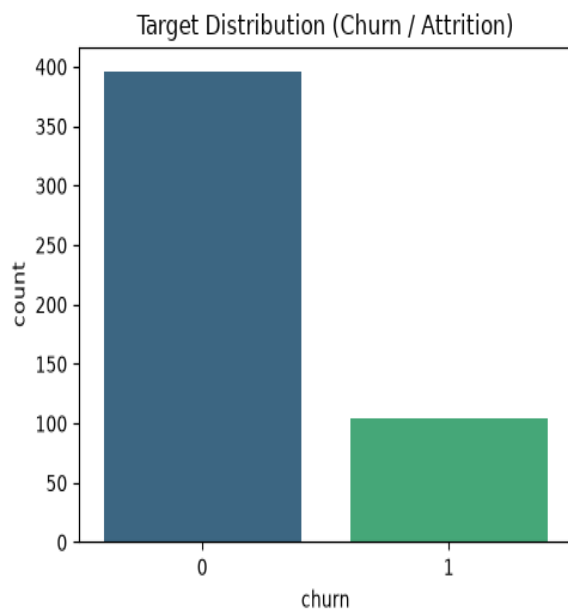
--- STEP 3: EXPLORATORY DATA ANALYSIS ---
   Age  MonthlyIncome  DistanceFromHome  YearsAtCompany  JobSatisfaction  OverTime  churn
count  496.000000    496.000000    496.000000    496.000000    496.000000    496.000000  496.000000
mean    38.778226    10342.645161    8.412997    6.863306    2.759881    0.506048    0.199597
std     9.132146    3723.950021    6.029581    3.705289    1.113895    0.500473    0.399574
min     22.000000    2501.000000    1.000000    1.000000    1.000000    0.000300    0.000000
25%     31.000000    7062.250000    4.000000    4.000000    2.000000    0.000000    0.000000
50%     38.000000    10066.500000    8.000000    7.000000    3.000000    1.000000    0.000000
75%     46.000000    13798.750000    13.000000    10.000000    4.000000    1.000000    0.000000
max     59.000000    19996.000000    29.000000    14.000000    4.000000    1.000000    1.000000
Saved 'experiment10/eda_plots.png' successfully.

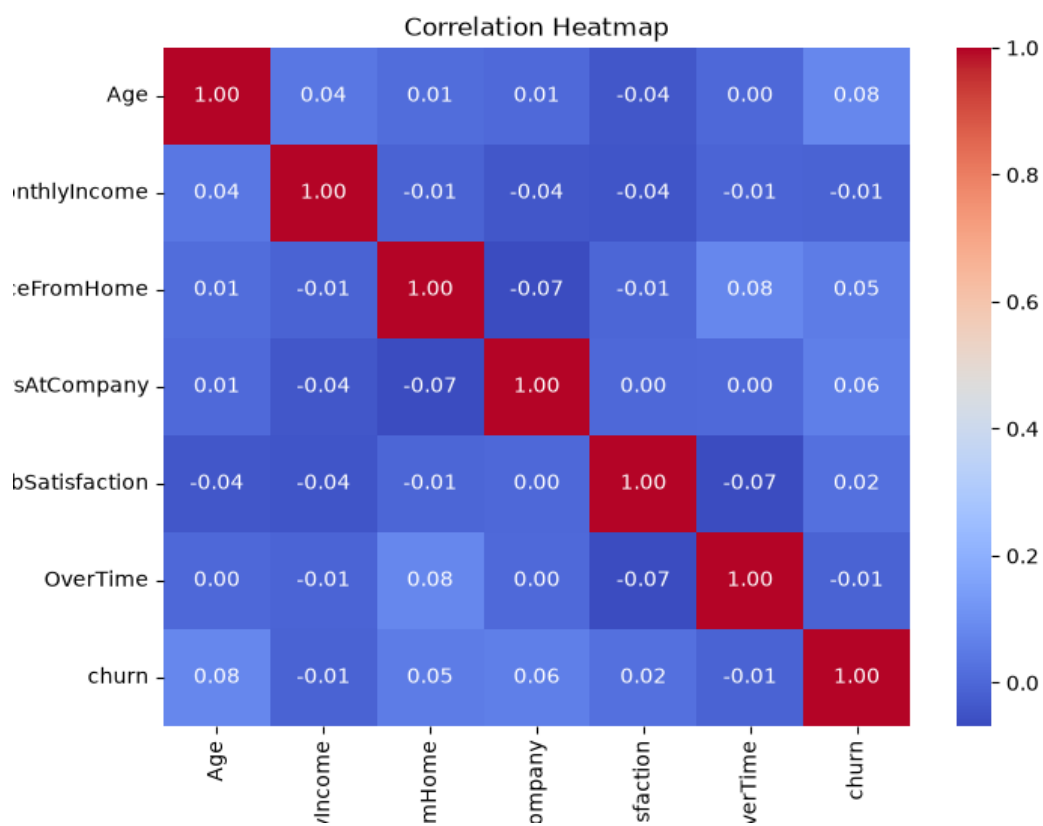
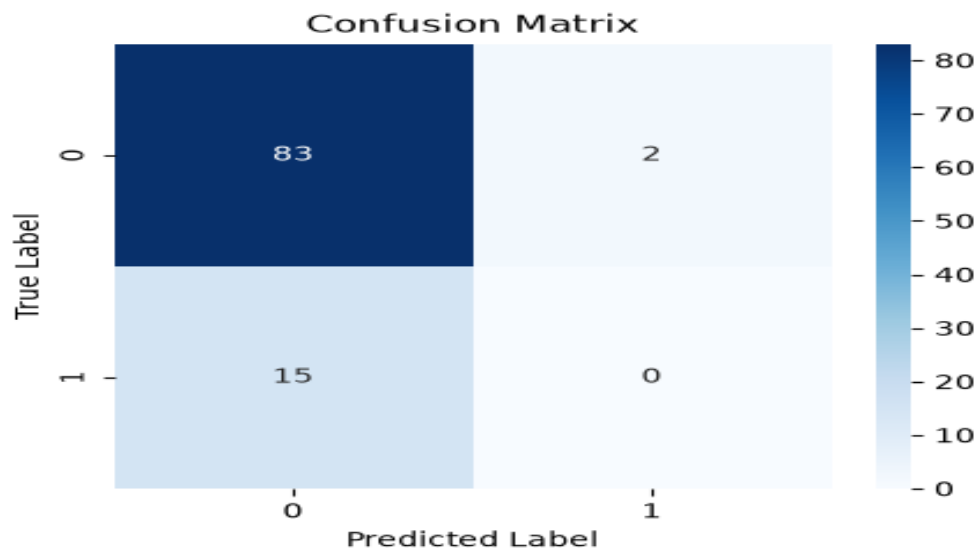
--- STEP 4: STATISTICAL ANALYSIS ---
Saved 'experiment10/correlation_heatmap.png' successfully.
Correlations with Target Variable (churn):
churn      1.000000
OverTime   0.206312
DistanceFromHome 0.092488
MonthlyIncome 0.079348
Age         0.061390
YearsAtCompany -0.042933
JobSatisfaction -0.102457

--- STEP 5 & 6: MODEL BUILDING ---
--- STEP 7: EVALUATION METRICS ---
Accuracy Score: 85.88789677419355 %
F1 Score: 0.8494268572116866
Detailed Classification Report:
   precision  recall  f1-score  support
0      0.88      0.95      0.91       80
1      0.75      0.50      0.60       16
accuracy    0.81    0.73    0.76       96
macro avg   0.81    0.73    0.76       96
weighted avg 0.85    0.86    0.85       96

Saved 'experiment10/confusion_matrix.png' successfully.
[SUCCESS] End-to-end data analytics project completed successfully!

```





Questions & Answers

Q1. What are the major stages involved in an end-to-end data analytics project?

- **Problem Definition:** Setting clear business goals and analytics objectives.
 - **Data Collection:** Importing relevant datasets from databases, APIs, or flat files.
 - **Data Preprocessing:** Cleaning data by handling missing values, duplicates, outliers, and encoding variables.
 - **Exploratory Data Analysis (EDA):** Discovering patterns and distributions through descriptive statistics and plots.
 - **Statistical Analysis:** Running correlation analysis and hypothesis testing to validate relationships.
 - **Predictive Modeling:** Building ML models (Regression, Classification, or Clustering) based on the problem statement.
 - **Model Evaluation:** Performance checking using metrics like Accuracy, F1-Score, RMSE, or MAE.
 - **Insights & Reporting:** Developing interactive BI dashboards and delivering actionable business strategies.
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Q2. Why is data preprocessing considered the foundation of successful analytics and machine learning?

- Follows the "Garbage In, Garbage Out" principle—flawed data yields unreliable models.
 - Missing values and wrong data types cause execution errors during model training.
 - Skewed distribution and unscaled features introduce bias toward larger values.
 - Clean data improves model learning efficiency and generalization on unseen data.
-

Q3. How does Exploratory Data Analysis (EDA) contribute to model development?

- Uncovers underlying feature distributions, variances, and skewness.
 - Detects multicollinearity and feature correlation to assist in Feature Selection.
 - Highlights anomalies, outliers, and data errors before modeling.
 - Guides feature engineering strategies to boost overall predictive performance.
-

Q4. What factors should be considered while selecting a machine learning algorithm for a business problem?

- **Problem Category:** Supervised vs. Unsupervised (Classification, Regression, Clustering).
 - **Dataset Size & Dimensionality:** High-dimensional data requires scalable or tree-based algorithms.
 - **Interpretability vs. Accuracy:** High-stake fields require explainable models (e.g., Decision Trees) over black-box models.
 - **Computational Cost & Latency:** Real-time predictions require low-latency algorithms.
-

Q5. Why is model evaluation essential before deploying a predictive model?

- Detects overfitting (high training accuracy, poor real-world performance) and underfitting.
 - Assesses real-world prediction stability on unseen validation test sets.
 - Minimizes business and financial risks tied to false predictions.
 - Establishes benchmark comparisons against baseline models.
-

Q6. Explain how Business Intelligence tools complement machine learning in data analytics projects.

- Converts complex numerical predictions into accessible dashboards (Tableau / Power BI).
 - Bridges the gap between technical data engineering and executive decision-making.
 - Enables real-time KPI tracking and interactive filtering for end-users.
-

Q7. What challenges are commonly encountered while working with real-world datasets?

- **Incomplete Data:** High proportion of missing, corrupted, or null values.
 - **Class Imbalance:** Target categories skewed towards a single class (e.g., fraud or churn).
 - **Noise and Outliers:** Uncleaned data points degrading model performance.
 - **Data Privacy and Governance:** Compliance constraints (e.g., GDPR, HIPAA).
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Q8. How can analytical insights be converted into actionable business recommendations?

- Mapping technical model metrics directly to business KPIs (e.g., lower churn = retained revenue).
 - Providing targeted action plans (e.g., offering loyalty discounts to high-risk churn customers).
 - Evaluating potential ROI and operational risks before implementation.
-

Q9. Why is effective presentation and data storytelling important in analytics projects?

- Translates technical machine learning concepts into clear strategic insights.
 - Builds trust and credibility with non-technical stakeholders and leadership.
 - Accelerates decision-making and budget approvals for analytics initiatives.
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Q10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution.

- **Advanced Algorithms:** Integrating Gradient Boosting models (XGBoost, LightGBM) or Deep Learning.
- **Automated MLOps Pipelines:** Setting up CI/CD workflows for automated retraining and model monitoring.
- **Domain Feature Engineering:** Constructing high-value domain-specific synthetic features.
- **Real-time Data Streams:** Transitioning from batch analysis to continuous real-time data processing.

